IGCSE 0607 Practice Questions — E4.1 — Combined Questions & Markscheme

Coordinates

Total Questions: 3 — **Total Marks:** 20 Marks

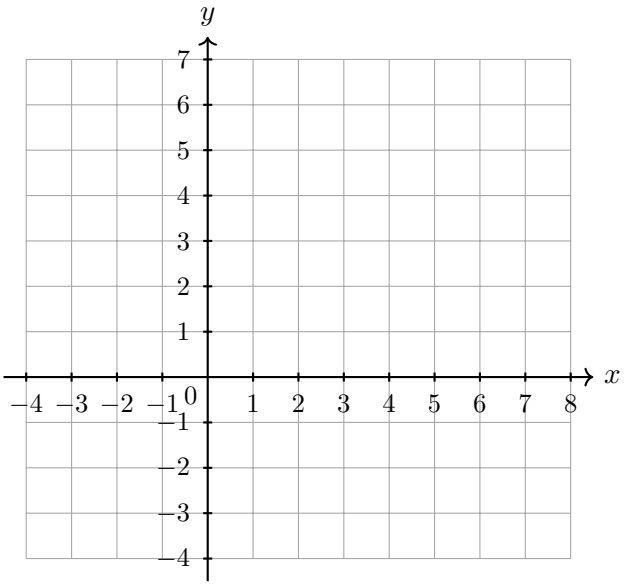
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  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(1 Q)*

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1 The points $A(-1,3)$, $B(2,-1)$ and $C(6,1)$ are three vertices of a parallelogram $ABCD$.

(a) On the grid, plot the points A, B and C.



[1]

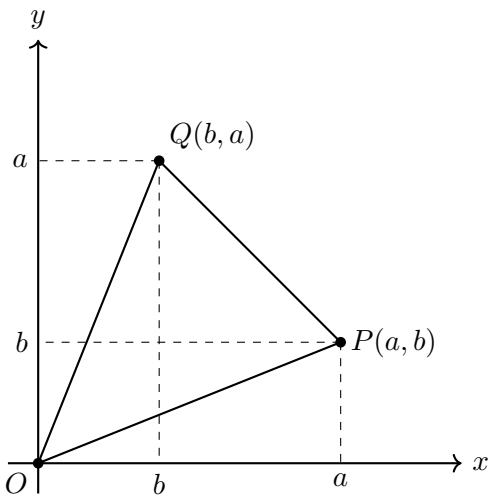
(b) Find the coordinates of the fourth vertex, D.

(..... , ...) **[2]**

(c) Write down the coordinates of the centre of the parallelogram.

(..... , ...) **[2]**

2 The points $P(a, b)$ and $Q(b, a)$ are in the first quadrant, with $a > b$.
The origin is denoted by $O(0, 0)$.



(a) By drawing a suitable rectangle or otherwise, show that the area of triangle OPQ is $\frac{1}{2}(a^2 - b^2)$.

[3]

(b) The area of triangle OPQ is 24 and the length of PQ is $4\sqrt{2}$.
Find the values of a and b.

$a =$ ..... , $b =$ ..... **[5]**

3 A straight line L has the equation $3x + 2y = 18$.

(a) The line L crosses the x-axis at point A and the y-axis at point B.
Find the coordinates of A and B.

$A =$

$B =$ **[2]**

(b) Point P lies on the line segment AB such that the ratio $AP : PB = 2 : 1$.
Find the coordinates of P.

(..... , ...) **[3]**

(c) Does the point $(4, 3)$ lie on the line L?
Show how you decide.

[2]

1 **(a)** All three points A, B, and C plotted accurately on the grid **[B1]**

(b) Identifies the horizontal and vertical translation, e.g. $\vec{BA} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ OR correctly plots D on the grid **[M1]**

$(3, 5)$ **[A1]**

(c) Calculates the midpoint of AC or BD: $\left(\frac{-1+6}{2}, \frac{3+1}{2}\right)$ **[M1]**

$(2.5, 2)$ **[A1]**

[Back to Question] *[Total Marks: 5]*

2 **(a)** Identifies the area of the enclosing square as a^2 and the three outer right-angled triangles as $\frac{1}{2}ab$, $\frac{1}{2}ab$, and $\frac{1}{2}(a-b)^2$ **[M1]**

Sets up the subtraction for the area of OPQ: Area $= a^2 - \frac{1}{2}ab - \frac{1}{2}ab - \frac{1}{2}(a-b)^2$ **[M1]**

Correctly expands brackets and simplifies algebraically to $\frac{1}{2}(a^2 - b^2)$ without errors **[A1]**

(b) Sets up the distance squared equation: $(a-b)^2 + (b-a)^2 = (4\sqrt{2})^2$

2(a-b)$^2 = 32$ [M1]

Solves to find $a - b = 4$ (accepting $a > b$) **[A1]**

Uses the area equation:

$$\frac{1}{2}(a^2 - b^2) = 24$$

a$^2 - b^2 = 48$ [B1]

Factorises to $(a-b)(a+b) = 48$, substitutes $a - b = 4$ to obtain $a + b = 12$ **[M1]**

$a = 8$ and $b = 4$ **[A1]**

[Back to Question] *[Total Marks: 8]*

3 **(a)** Substitutes $y = 0$ to find $A(6, 0)$ **[B1]**

Substitutes $x = 0$ to find $B(0, 9)$ **[B1]**

(b) Recognises the change in x is -6 and change in y is 9 from A to B **[M1]**

Forms a valid ratio equation for either coordinate, e.g. x-coordinate $= 6 + \frac{2}{3}(-6)$ OR y-coordinate $= 0 + \frac{2}{3}(9)$ **[M1]**

$(2, 6)$ **[A1]**

(c) Substitutes $x = 4$ and $y = 3$ into the equation: $3(4) + 2(3)$ **[M1]**

Shows $12 + 6 = 18$ and clearly concludes 'Yes' **[A1]**

[Back to Question] *[Total Marks: 7]*